

Scores as Processes

A graded process calculus for composition and algorithmic music:
notes as communication, idioms as formulae, style and physicality as
grading

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Second version. Companion to *Graded Where-Clauses over the Reals* (Paper II); the idea
generalises the F1R3Skein instrument

Abstract

Conventional notation writes down one performance. It has no native way to say who is listening to whom, which of several continuations a style prefers, or what lies easily under a player's hands; those things are left to the performer's training. This note proposes a notation in which they are the primary objects. A score is a process in a variant of the rho calculus, and a performance is a run of it.

Names are decorated with a timbre and a note datum. A name carrying a pitch is a *channel*, a name carrying a duration is a *co-channel*, and communication happens only between the two, so every reduction emits exactly one note and there are no silent steps. The musical vocabulary then reads off parallel composition. *Melody* arises from interaction at composition, where each note sets up the next through its continuation; *harmony* arises from independence at composition, where events commute and notes at equal musical times sound together; *choice* arises from contention. Because the medium is message passing, idioms built on exchange — call and response, trading phrases, interlocking parts, cross-rhythm — are written directly as the communication structure of the score rather than approximated on a grid.

Receipts carry the graded where-clauses of Paper II, and the clause does two jobs. Its *grading* gives distributions: independent factors for style, for register, and for the physicality of an instrument multiply, and after normalisation they yield exactly the product of the factors' preferences, so a style can be moved to a new instrument by exchanging one factor, or a new physicality imagined by writing one no human has. Its *formulae* carry motifs and idioms: spatial modalities read the voice's own past, which the calculus keeps in the voice's location, and context-labelled modalities read what a choice leads to in the company of the other players. Crisp formulae enforce an idiom, graded ones encourage it, and a fixed-point formula guarantees that enforcement never paints a voice into a corner. This is the dance between freedom and constraint that jazz makes its own: a score contains non-determinism, an improvised performance is a trace through it, and the crisp layer fixes which traces are admissible while the grading says which the idiom favours.

Results: every reduction is a note; a note cannot be fitted from alternatives on both sides at once, which forces a single *key* and one datum chosen a note early; a voice is a key and a complete fan at a location that quotes its own past, so unplayed alternatives are inert; unbounded voices need no recursion former, because reflection supplies it and its communications are null notes, rests of length zero, which are silent and take no time; the law of a voice is exactly the normalised product of its clause's factors; independent voices have product law and interleaving-invariant performances; and handing a line from one player to another hands over its history, so the responder can read the call. The traditional division of musician and instrument falls out of the polarities: keys belong to instruments and touches to players, a $2n$ -hands piano is a few lines of process, synchronous output lets

a player write a phrase, and an instrument whose keys hold several timbres chooses its timbre at every touch from how it is played. A stdlib-only reference simulator checks the quantitative claims.

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1 Introduction

1.1 What a score could say

A page of conventional notation is a very good description of one performance. It fixes pitches and durations on a shared grid, and it leaves to the performer almost everything that makes the music idiomatic: which note a line would *rather* go to, what a phrase is answering, what a hand can reach. Where the music is itself a structure of exchange — the trading of phrases in jazz, the call and response and interlocking parts found across many African musics — the grid is a poor fit, because the thing that matters is who plays in response to what, and the grid records only the result.

This note proposes a notation in which three things are written explicitly.

1. **Who plays with whom.** A score is a process. Players are subprocesses, and the ways they interact are the communications between them. Handing a line from one player to another is a communication; so is alternating notes of one line between two players.
2. **How likely each continuation is.** Every place where a player chooses what to play next carries a *grading*, a nonnegative weight on each alternative. Weights compose: a factor for the style, a factor for the register, a factor for the instrument's physicality. Normalising the product gives the distribution the score actually plays.
3. **What the music must or should do.** The grading is attached by a formula, and formulae can inspect structure. Spatial modalities read the recent past, so motifs can be recognised and completed; context-labelled modalities read what a choice leads to in the company of the other players, so idioms can be enforced without deadlock.

A score written this way is a distribution over performances, which makes it a direct specification for algorithmic generation. It is equally a compositional instrument: a composer can hold the communication structure fixed and exchange the style factor, keep a style and exchange the physicality, or keep everything and harden one idiom from a preference into a rule.

Freedom and constraint. Much of the appeal of jazz lies in the dance between freedom and constraint: the player is free, but free within a tune, a set of changes, a style, and what the band is doing. Conventional notation has no place for that dance. A page either fixes the notes, leaving nothing to the player, or writes chord symbols and leaves the constraint to convention. A score in this calculus can say both things at once. It contains non-determinism, and an improvised performance is a *trace* through it: one path among the paths the score allows. The score does not merely permit improvisation; it says which traces are admissible and which are not, and, among the admissible ones, which the idiom favours. Constrained non-determinism is the native form of such a score, not an approximation to it (§7.3).

1.2 What composition can do

The key design decision is that a note is the only thing a communication can produce. Names are decorated so that each is either a channel, carrying a pitch, or a co-channel, carrying a duration, and the only communication rules pair a channel with a co-channel. A communication is then a note, the pitch coming from one side and the duration from the other. Once that holds, the musical vocabulary is a question about what parallel composition puts in contact:

what the composed parts share	what happens at composition	music
no active location	independence: events commute	harmony
a location, one way to meet	interaction: the next note is set up	melody
a location, several ways to meet	contention: some must lose	choice

Melody and harmony both come from parallel composition; they differ by whether the composed parts interact or are independent. The third row is where grading lives. These are the grades that the graded-where analysis found for overlap in general: inert, communicating and contending [2].

1.3 Background: the F1R3Skein instrument

The calculus grew out of F1R3Skein, a virtual instrument that draws a pitch stream and a duration stream — originally from the digits of transcendental constants, later from state machines the player draws — and fits them into notes [5, 6]. The instrument’s picture of two separately generated streams meeting at a note is what suggested channels and co-channels. Its aims are different: it is an instrument for a performer, and this calculus is a notation for a composer. The instrument’s streams are expressible as particular scores (Remark 6.3), but nothing below depends on it.

1.4 Contributions

1. The *score calculus* (§2), whose names may leave their decorations open so that the **rho** round trip holds as an instance relation, $x \sqsubseteq @*x$ (Proposition 2.5), illustrated on *Twinkle, Twinkle, Little Star* (§2.5), in which every reduction emits exactly one note (Proposition 2.10), with a timed semantics in which musical time is carried by the data, and a proof that independent parts perform the same under every interleaving (Theorem 2.14).
2. The *graded layer* (§3): where-clauses whose formulae include spatial and context-labelled modalities and whose values are weights, resolved by normalisation within contention sets.
3. The *one-sided fitting* result (Proposition 4.1): a note cannot be chosen from alternatives on both sides, which forces a single key and one datum chosen a note early.
4. The *voice* (§5): a key and a complete fan at a location that quotes the voice’s past; unplayed alternatives are inert (Lemma 5.6), and the past is readable by formulae. Unbounded voices are built by reflection alone, through a self-refreshing server whose communications are null notes (Proposition 5.3).
5. *Distributions* (§6): the law of a voice is the normalised product of its factors (Theorem 6.1); product of factors and convolution of interval laws; style and physicality as separable factors; chords as joins, which is where vertical physicality lives.
6. *Motifs and idioms* (§7): recognising and completing motifs with spatial formulae; enforcing idioms with crisp formulae, and keeping enforcement live with a greatest fixed point (Proposition 7.2); improvisation as a trace through a score whose crisp layer fixes which traces are admissible and whose grading fixes how likely they are (Proposition 7.4); message-passing idioms as score structure (§7.4).
7. *Harmony* (§8): independent voices have product law and schedule-invariant performances (Theorem 8.1); shared locations couple voices instead (Proposition 8.2).
8. *Musicians and instruments* (§9): keys belong to instruments and touches to players; keys re-arm themselves by reflection and sound once at a time (Proposition 9.2); synchronous output sequences a player’s phrase exactly (Proposition 9.4); a $2n$ -hands piano as a worked

example; and a chimeric instrument whose timbre is chosen at every touch by how it is played (Proposition 9.6).

9. A reference simulator, `skeinsim.py`, whose measurements are in §10, including the check that the reflective voice plays exactly the notes of the step semantics.

1.5 Scope

Tempo is out of scope; it is the natural second grading dimension and the last section says where it would attach. Dynamics and articulation are further decorations of the same kind and are not treated. Learning a style from a corpus, which Paper II's gradient makes available, is not attempted here.

2 The score calculus

2.1 Data, names and processes

A score fixes a finite *pitch alphabet* Π containing a rest r , and a finite *duration alphabet* Δ together with a length $|\cdot| : \Delta \rightarrow \mathbb{Q}_{\geq 0}$ measured in whole notes. The duration alphabet is the composer's: it may include dotted values and tuplets, and cross-rhythms such as three against two need it to (Example 7.6). It always contains a *null duration* ϵ with $|\epsilon| = 0$, and we write Δ^+ for the durations of positive length. Let $\mathcal{T}$ be a set of timbres. We write π for pitch data, δ for duration data, and $d \in \Pi \uplus \Delta$ for a note datum of either kind.

Definition 2.1 (Syntax). Names and processes are given by

$$\begin{aligned} \hat{\tau} &::= \tau \mid _ & \hat{d} &::= d \mid _ \\ x &::= y \mid \langle @P, \hat{\tau}, \hat{d} \rangle \\ P, Q &::= 0 \mid \text{for}(y \leftarrow x) P \mid x!(Q, \hat{\tau}, \hat{d}) \mid P \mid Q \mid *x \end{aligned}$$

where y ranges over name variables, $\tau \in \mathcal{T}$, $d \in \Pi \uplus \Delta$, and $_$ is the *wildcard*. The receipt $\text{for}(y \leftarrow x) P$ binds y in P . The *polarity* of a closed name is fixed by its datum: $\langle @P, \hat{\tau}, \pi \rangle$ is a *channel*, $\langle @P, \hat{\tau}, \delta \rangle$ is a *co-channel*, and a name whose datum is $_$ has no polarity. We write $@P$ for the *general name* $\langle @P, _, _ \rangle$; in particular $@*x = \langle @*x, _, _ \rangle$ is a name. The overloading is deliberate: the general name at a location is the name that says nothing but the location.

A name is a triple: a quoted process, its *location*, and a decoration consisting of a timbre and a datum, either of which may be left open. A message $x!(Q, \hat{\tau}, \hat{d})$ carries a process Q and a decoration of its own, from which the receiver's new name is built. A receipt on a channel holds a pitch and waits; we call it a *key*. A message on a co-channel holds a duration; we call it a *touch*. The names anticipate §9, where keys belong to instruments and touches to players.

Remark 2.2 (Recursion is reflective). There is no replication and no recursion former. As in rho, recursion comes from reflection: a process receives code as a quoted process and runs it with $*$ [7]. Here the communications that do this are notes like any other, and §5.2 arranges for them to be *null notes*, rests of length zero, so that the machinery of recursion is silent and takes no time. A piece of bounded length needs none of it: it is a finite term.

2.2 Equivalence and reduction

Definition 2.3 (Structural congruence and name equivalence). Structural congruence $\equiv$ is the least congruence, closed under α -conversion, such that $(\mid, 0)$ is a commutative monoid and

$$*\langle @P, \hat{\tau}, \hat{d} \rangle \equiv P.$$

Two locations are equivalent, $@P =_{\mathbf{N}} @P'$, when $P \equiv P'$; two names are equivalent when their locations are and their decorations are equal, a wildcard being equal only to a wildcard. Names inside processes are compared up to this equivalence.

Dropping a name forgets its decoration, so in the absence of wildcards there would be no way back from $*x$ to a name, and the ρ law $@*x = x$ would not even be well typed. With wildcards there is a way back, but it is an instance relation, not an equation.

Definition 2.4 (Matching). Two decoration components *match* when they are equal or one of them is $_$; their *meet* $\hat{a} \sqcap \hat{b}$ is then the concrete one if either is concrete, and $_$ otherwise. A name x is an *instance* of x' , written $x \sqsubseteq x'$, when their locations are equivalent and each decoration component of x' is either $_$ or equal to that of x .

Proposition 2.5 (The round trip). *For every closed name x : $x \sqsubseteq @*x$; the general name $@*x$ is the greatest name at x 's location in the instance order; and $*@*x \equiv *x$.*

Proof. If $x = \langle @P, \hat{\tau}, \hat{d} \rangle$ then $*x \equiv P$, so $@*x = \langle @*x, _, _ \rangle$ has location equivalent to $@P$ and an all-wildcard decoration, of which every decoration is an instance. Dropping it gives $*x$ again. $\square$

Remark 2.6 (Why not an equation). Adding $@*x =_{\mathbf{N}} x$ to name equivalence would identify $\langle @P, \tau, \pi \rangle$ and $\langle @P, \tau', \delta \rangle$ with $@P$, and hence with each other: every channel and co-channel at a location, in every timbre, would collapse into one name. Matching keeps the decorations distinct while letting a general name stand for all of them. It is used by the communication rules and by nothing else; structural congruence is unchanged.

Definition 2.7 (Reduction). Reduction is the least relation closed under

$$\begin{array}{l}
\text{COMM}_1 \quad \frac{@S =_{\mathbf{N}} @T \quad \hat{\tau}_1 \sqcap \hat{\tau}_2 = \tau}{\text{for}(y \leftarrow \langle @S, \hat{\tau}_1, \pi \rangle) P \mid \langle @T, \hat{\tau}_2, \delta \rangle!(Q, \hat{\tau}_3, \hat{d}) \rightarrow P\{\langle @Q, \hat{\tau}_3, \hat{d} \rangle/y\}} \\
\text{COMM}_2 \quad \frac{@S =_{\mathbf{N}} @T \quad \hat{\tau}_1 \sqcap \hat{\tau}_2 = \tau}{\text{for}(y \leftarrow \langle @S, \hat{\tau}_1, \delta \rangle) P \mid \langle @T, \hat{\tau}_2, \pi \rangle!(Q, \hat{\tau}_3, \hat{d}) \rightarrow P\{\langle @Q, \hat{\tau}_3, \hat{d} \rangle/y\}} \\
\text{PAR} \quad \frac{P \rightarrow P'}{P \mid R \rightarrow P' \mid R} \quad \text{STRUCT} \quad \frac{P \equiv P_1 \quad P_1 \rightarrow P'_1 \quad P'_1 \equiv P'}{P \rightarrow P'}
\end{array}$$

Matching concerns the two *subjects* only. Their locations must be equivalent and their timbres must meet in a concrete timbre, which is the note's; a receipt on $\langle @L, _, \pi \rangle$ listens in any timbre and sounds in the one the sender addressed. Their data are free to differ, and it is their difference that makes the note: the receipt's datum and the message subject's datum must be concrete and of opposite polarity, since a wildcard supplies neither a pitch nor a duration.

The *payload* takes no part in matching. Its process and its decoration $(\hat{\tau}_3, \hat{d})$ are data, carried across the communication unchanged to form the name the receiver binds; either component may be concrete or $_$. In particular a message $x!(Q, _, _)$ passes exactly the general name $@Q$, and a message whose payload names a timbre other than the note's hands the receiver a name in that other timbre. The note sounds in the timbre the two subjects agree on; what happens next may be in another.

The rules are not labelled. Nothing needs to be added to a reduction to know what it played, because the communication that effects it already contains the answer.

Definition 2.8 (Record of a reduction; its note). Every derivation of $P \rightarrow P'$ has exactly one instance of COMM_1 or COMM_2 at its leaf. The *record* of the reduction is the data that

instance matched: the receipt's subject $\langle @S, \hat{\tau}_1, d_1 \rangle$, the message's subject $\langle @T, \hat{\tau}_2, d_2 \rangle$, and the payload $(Q, \hat{\tau}_3, \hat{d})$. Its *note* is

$$\nu(\text{record}) := (\hat{\tau}_1 \sqcap \hat{\tau}_2, \pi, \delta),$$

where π is whichever of d_1, d_2 is a pitch and δ whichever is a duration. The *playback* of a run is the sequence of the notes of its records.

A record carries more than is needed to play the note — the locations, the payload, which side listened — but it always carries what is needed, and it is exactly what had to be present for the communication to happen at all. Playback reads it off; it is not a label the semantics must maintain.

Remark 2.9 (Open systems). In an open system, where transitions are labelled by the contexts that enable them, $P \xrightarrow{K} P'$, the label K supplies the half of the redex that P lacks. The label together with the half that P contributes is again a record, so playback needs nothing beyond the context-labelled transition system the calculus already has.

Proposition 2.10 (Every reduction is a note). *(i) Every reduction has a record, and every record has a note. There are no silent steps.*

(ii) A receipt and a message whose subjects have the same polarity never communicate, and neither does a subject with no polarity; nor do two subjects whose timbres are both $_$.

(iii) The note depends only on the two data, not on which side listens: COMM_2 is COMM_1 with the polarities exchanged.

Proof. (i) PAR and STRUCT have one reduction premise each and the only axioms are the communication rules, whose premises make the subjects' timbres meet concretely and make one subject's datum a pitch and the other's a duration, so ν is defined on every record. (ii) and (iii) by inspection. $\square$

A rest is a pitch like any other, so a rest is a note: silence for the duration drawn. A note of length zero, in particular the *null note* (τ, r, ϵ) , is a note in every formal sense and sounds nothing for no time. It plays the part that silent internal steps play in other calculi, but it is not a separate kind of step: the calculus still has only notes.

2.3 Musical time

No reduction takes time; a note *occupies* time.

Definition 2.11 (Timed configurations and performance). A *timed configuration* stamps every top-level receipt and message with a time in $\mathbb{Q}_{\geq 0}$. A communication between a receipt stamped t_r and a message stamped t_m has *onset* $t = \max(t_r, t_m)$: a player cannot answer before it has been asked, nor play before it is ready. If the note of its record is (τ, π, δ) , its *timed note* is (t, τ, π, δ) , and it stamps the receipts and messages its continuation brings to top level with $t + |\delta|$. The initial configuration is stamped by the score (usually 0; an entrance is a later stamp). A run is *onset-ordered* if every communication has minimal onset among the enabled ones. The *performance* of a finite run is the multiset of its timed notes of positive length. Two timed notes *strike together* when their onsets are equal and *sound together* when their intervals $[t, t + |\delta|)$ overlap.

Remark 2.12 (Time in the data). In the weighted-GSLT note, time is dynamical, an exponential holding time in the style of Gillespie [4, 10]. None of that is used here. Paper II separates the embedded jump chain, which normalised selection defines, from holding times, which it does not [1]; a score is a jump chain with a clock read off the data. That is what leaves tempo free to be graded separately later.

2.4 Independence and interleaving

Definition 2.13 (Active locations; independence). The *active locations* $\text{act}(P)$ are the (location, timbre) pairs of the subjects of top-level receipts and messages, the timbre possibly $_$. P_1 and P_2 are *independent* if every configuration reachable from $P_1 \mid P_2$ is congruent to $C_1 \mid C_2$ with C_i reachable from P_i and no entry of $\text{act}(C_1)$ matching an entry of $\text{act}(C_2)$: equivalent locations with matching timbres.

Theorem 2.14 (Interleaving invariance). *Let P_1, P_2 be independent and fix the choices each makes when it fires. Then every onset-ordered run of $P_1 \mid P_2$ making those choices has the same performance, the union of the performances of P_1 and P_2 run separately.*

Proof. By independence every communication belongs to one component and consumes only its resources, so communications of different components commute, forming diamonds with the same records. A communication’s onset is the larger of its two stamps, both fixed by earlier communications of the same component from the durations that component emitted. So a component’s timed notes do not depend on how its events interleave with the other’s, and every onset-ordered interleaving is a permutation of the same two histories. $\square$

Remark 2.15 (Harmony needs no global step). “At the same time” does not require maximal progress between independent parts: simultaneity is equality of onsets, and onsets are intrinsic to each part’s own history. Onset ordering is maximal progress in musical time, and it matters only where parts contend.

2.5 Example: *Twinkle, Twinkle, Little Star*

Before any grading, the calculus can already write down a fixed piece. Take the first phrase of the tune — “Twinkle, twinkle, little star, how I wonder what you are” — with a plain bass line under it:

	bar 1	bar 2	bar 3	bar 4
melody	C4 C4 G4 G4	A4 A4 G4 (half)	F4 F4 E4 E4	D4 D4 C4 (half)
bass	C3 (half) C3 (half)	F3 (half) C3 (half)	F3 (half) C3 (half)	G3 (half) C3 (half)

Unmarked melody notes are quarters. The pitch alphabet contains the nine pitches used and the rest r ; the duration alphabet contains the quarter q with $|q| = \frac{1}{4}$, the half h with $|h| = \frac{1}{2}$, and the null duration.

One written note. A note of pitch p and duration d , played at a location L in timbre τ and followed by P , is a key on the channel for p together with a touch on the co-channel for d :

$$N_{L,\tau}(p, d; P) := \text{for}(y \leftarrow \langle @L, \tau, p \rangle) P \mid \langle @L, \tau, d \rangle!(0, _, _) \quad (y \text{ not free in } P).$$

The two meet by COMM_1 and emit (τ, p, d) . The name passed to the continuation is not used, so the message passes the general name $@0$ and leaves its decoration open.

The score. A part is a right-nested chain of written notes, and the piece is the parallel composition of its parts. With L_m and L_b two inequivalent locations,

$$\begin{aligned} \text{Melody} &:= N_{L_m, \text{piano}}(C4, q; N_{L_m, \text{piano}}(C4, q; N_{L_m, \text{piano}}(G4, q; \dots N_{L_m, \text{piano}}(C4, h; 0) \dots)) \\ \text{Bass} &:= N_{L_b, \text{bass}}(C3, h; N_{L_b, \text{bass}}(C3, h; N_{L_b, \text{bass}}(F3, h; \dots N_{L_b, \text{bass}}(C3, h; 0) \dots)) \\ \text{Twinkle} &:= \text{Melody} \mid \text{Bass} \end{aligned}$$

with fourteen written notes in the melody and eight in the bass, in the order of the table. It is a finite term: no recursion, no where-clauses, no grading.

Melody is interaction. At the start, the only top-level parts of **Melody** are the key holding C4 and the message holding a quarter, both at L_m and both stamped 0. They communicate, emitting (piano, C4, q) at onset 0, and the continuation brings the next key and message to top level stamped $\frac{1}{4}$. Each note sets up the next. Because only one key and one message of the melody are ever at top level, every contention set has exactly one candidate, and the melody can use a single location for all fourteen notes.

Harmony is independence. The bass lives at another location in another timbre, so the two parts share no active location and are independent. By Theorem 2.14 every onset-ordered run has the same performance:

onset	melody	bass	onset	melody	bass
0	C4 q	C3 h	2	F4 q	F3 h
$\frac{1}{4}$	C4 q		$\frac{9}{4}$	F4 q	
$\frac{1}{2}$	G4 q	C3 h	$\frac{5}{2}$	E4 q	C3 h
$\frac{3}{4}$	G4 q		$\frac{11}{4}$	E4 q	
1	A4 q	F3 h	3	D4 q	G3 h
$\frac{5}{4}$	A4 q		$\frac{13}{4}$	D4 q	
$\frac{3}{2}$	G4 h	C3 h	$\frac{7}{2}$	C4 h	C3 h

The eight rows in which both parts strike are the chords of the phrase, I I IV I IV I V I. Nothing in the term coordinates them. Each part keeps its own clock in its own durations, and the notes line up because the durations add up. The reference simulator plays this term under both tie-breaking orders and obtains the same twenty-two timed notes, ending at 4 whole notes.

What would go wrong. Write the bass at the melody’s location and in the melody’s timbre, and the parts are no longer independent. At onset 0 the melody’s key holding C4 could then meet the bass’s half-note message as well as its own quarter, and the two parts would trade durations. Independence is a property of (location, timbre) pairs, and a composer who wants independent parts keeps them apart. Proposition 4.1 makes precise what happens when they are not.

From a tune to a distribution of tunes. Every fitting in Twinkle has exactly one candidate, so the piece is the degenerate case of everything that follows. Replace each written message by a fan and each key by a graded key whose clause weights the tune’s own transitions heavily and others lightly, and the same score becomes a distribution over variations of the tune (§5 and §6); give its clause the phrase as a motif, and the variations can be required to return to it (§7).

3 The graded layer

3.1 Where-clauses, and what their formulae can see

Receipts acquire a clause, the clause-free form abbreviating $\lceil \top \rceil$:

$$P ::= \dots \mid \text{for}(y \leftarrow x \text{ where } \psi) P.$$

Clauses are drawn from the two formula sorts of Paper II [1]. The *crisp* sort β has full propositional structure and the modalities of the generated logic; the *graded* sort takes values in the resolution algebra $\mathbb{V} = (\mathbb{R}_{\geq 0}, +, \times, 0, 1)$:

$$\begin{aligned} \beta &::= \top \mid \alpha \mid \beta \wedge \beta \mid \neg \beta \mid \beta \parallel \beta \mid \text{out}(n)\beta \mid @\beta \mid \langle K \rangle \beta \mid \nu X.\beta \mid X \\ \psi &::= \lceil \beta \rceil \mid w \mid \psi \otimes \psi \mid \psi \oplus \psi \mid \langle K \rangle_{\mathbb{V}} \psi \mid \mu X.\psi \mid \nu X.\psi \mid X \end{aligned} \quad (w \in \mathbb{R}_{\geq 0})$$

where $\lceil \beta \rceil$ is 1 where β holds and 0 elsewhere, and α ranges over the atoms of Definition 3.1. The structural connectives — separating conjunction $\beta \parallel \beta$ (the formula counterpart of $P \mid Q$), output $\text{out}(n)\beta$, and the name predicate $@\beta$ that looks inside a quoted location — are the *spatial* modalities; $\langle K \rangle$, one per rewrite rule and one-hole context K , is the *context-labelled* behavioural modality: $\langle K \rangle \beta$ holds of a process R when $K[R]$ can take a step to a process satisfying β [8, 9, 3]. The graded $\langle K \rangle_{\mathbb{V}}$ sums over those steps, as in Paper II’s transfer operator.

What makes the formulae musically useful is *what they are evaluated on*.

Definition 3.1 (Candidates and their views). A *candidate* is a pair $c = (r, m)$ of a top-level receipt and message meeting the premises of COMM_1 or COMM_2 . A clause is evaluated on c *against the configuration in which it stands*, and sees:

- the note $\nu(c) = (\tau, \pi(c), \delta(c))$ it would emit, through the atoms $\text{pitch}(s)$ and $\text{dur}(v)$;
- the datum $\kappa(c)$ the message carries forward, through $\text{carry}(e)$, and relations between data such as $\text{step}(k) : \Leftrightarrow \kappa(c) - \pi(c) = k$ for ordered pitch alphabets;
- the location $@S$ at which it meets, through spatial formulae under the name predicate: $\text{at}(\beta) : \Leftrightarrow S \models \beta$;
- the name it would pass, $\langle @Q, \hat{\tau}, \hat{d} \rangle$, likewise: $\text{passes}(\beta) : \Leftrightarrow Q \models \beta$;
- the continuation it would produce, placed in a context: behavioural formulae $\langle K \rangle \beta$ are evaluated on $K[P']$ where P' is the residue of firing c .

Two uses follow and will be developed separately. The *grading* decides how much each candidate is preferred (§6); the *formulae* decide which structures in the past, and which futures in company, earn or forfeit that preference (§7).

3.2 Weighted communication and resolution

Definition 3.2 (Weighted communication). If c is a candidate at a receipt with clause ψ and $\psi \llbracket c \rrbracket > 0$, then c offers the reduction of Definition 2.7 with weight $\psi \llbracket c \rrbracket$. A candidate of weight 0 offers nothing.

Two candidates *contend* when they share a receipt or a message. A *contention set* is a connected component of contention among positive-weight candidates; a *matching* is a set of pairwise non-contending candidates of one set, *maximal* if nothing can be added.

Definition 3.3 (Resolution). A contention set is *resolved* by selecting a maximal matching $\mathcal{M}$ with probability proportional to $\prod_{c \in \mathcal{M}} \psi_c \llbracket c \rrbracket$ and firing all its candidates. Distinct contention sets are resolved with independent chance, in onset order, a set’s onset being the least onset of its candidates.

This is Paper II’s completion with maximal matchings as the alternatives [1, 2]. The chance source is a parameter: a pseudo-random generator, a physical source, or the digits of a constant. It may equally be a person: a player choosing, at each fitting, among the candidates of positive weight (§7.3). A score together with a fixed source is a single determined piece.

Proposition 3.4 (Weights are relative). *In a contention set whose maximal matchings all have the same size — any star in particular — multiplying every candidate’s value by the same positive constant leaves the resolution law unchanged.*

Proof. A matching of size k scales by c^k ; with one size throughout, numerator and denominator scale alike. $\square$

So a composer writes weights on whatever scale is convenient; only ratios are heard.

4 A note cannot be fitted from both sides

The obvious way to let both the pitch and the duration of the next note vary is to put pitch alternatives on one side of a location and duration alternatives on the other. It does not produce one note.

Proposition 4.1 (Two-sided fitting). *At one location let there be $m \geq 1$ receipts on channels with distinct pitches and $n \geq 1$ messages on co-channels with distinct durations, every candidate of positive weight. They form one contention set whose maximal matchings all have $\min(m, n)$ candidates. Hence (i) one resolution emits $\min(m, n)$ notes at once, drawn without replacement from both sides; and (ii) firing one candidate at a time, the leftovers go on to emit $\min(m, n) - 1$ more. A single note with both data chosen from alternatives requires $\min(m, n) = 1$.*

Proof. The candidates are the edges of $K_{m, n}$, whose line graph is connected; a matching of size below $\min(m, n)$ leaves a receipt and a message unmatched, and their edge extends it. $\square$

Corollary 4.2 (The key). *In a fitting that emits exactly one note, one side is a single receipt — the key — or a single message. With a key, the key’s own datum is fixed before the fitting, which chooses among messages, that is, among pairs (the other datum, the carried datum). One of the two note data is chosen a note early and carried forward.*

Musically, the key is already down: the pitch was decided when the key was chosen, and the fitting decides how long it sounds and what comes next. The look-ahead costs nothing in law (Theorem 6.1) but it is forced.

5 The voice

A voice is a single melodic line. We give it in two layers: the *step*, a finite term that plays one note, and the *server*, which uses reflection to supply the next step, so that a voice can run without bound. A piece of written length k can dispense with the server and write its k steps out.

5.1 The step: a key and a fan

At each note the line has a location, the quote of its past. At that location sits a *fan* of messages, one for every duration and every carried pitch, and a *key* holding the pitch the line has already chosen. Each message’s payload is a record of the note it would make, and the quote of that record becomes the next location.

Definition 5.1 (Record, fan, seed). Fix a timbre τ .

$$\begin{aligned} \text{Rec}(x, v, t) &:= \langle @*x, \tau, v \rangle!(*x, \tau, t) \\ \text{Fan}(x) &:= \prod_{v \in \Delta^+} \prod_{t \in \Pi} \langle @*x, \tau, v \rangle!(\text{Rec}(x, v, t), \tau, t) \\ H_0 &:= \langle @B, \tau, u_0 \rangle!(B, \tau, p_0) \end{aligned}$$

where B is a *base*, a closed receipt on a dead channel such as $\text{for}(y \leftarrow \langle @0, \tau_\perp, r \rangle) 0$ with $\tau_\perp$ used by no message. The seed record H_0 stands for a virtual note before the first: u_0 plays the role of the previous duration and p_0 is the first pitch.

When $x = \langle @H, \tau, p \rangle$, $*x \equiv H$ and $\text{Rec}(x, v, t) \equiv \langle @H, \tau, v \rangle!(H, \tau, t)$: “after the past H , the duration v was played and the pitch t comes next”. Only the location of x is used, never its datum.

The key must hold the pitch recorded at the location. That pitch is a runtime datum, and a received name gets its datum from the message that delivered it, which can only write data literally. So the step does not try to reconstruct the right key; it offers every key and lets the clause keep the right one:

Definition 5.2 (Step). For a clause ψ and a continuation $K(z)$,

$$\text{Step}_K(w) := \prod_{t \in \Pi} \text{for}(z \leftarrow \langle @*w, \tau, t \rangle \text{ where } \psi \otimes [\text{held}(t)] \mid K(z) \mid \text{Fan}(w),$$

where $\text{held}(t) := \text{at}(\langle -, -, - \rangle! (\top, -, t))$ holds when the location's record carries t . When the clause and timbre need naming we write $\text{Step}_K^{\psi, \tau}$.

At a location whose record carries p , the key holding p is the *live key*; the others have a crisp factor that is false at that location and always will be, so they have weight 0 on every candidate and offer nothing. Every candidate of the live key is a COMM_1 instance.

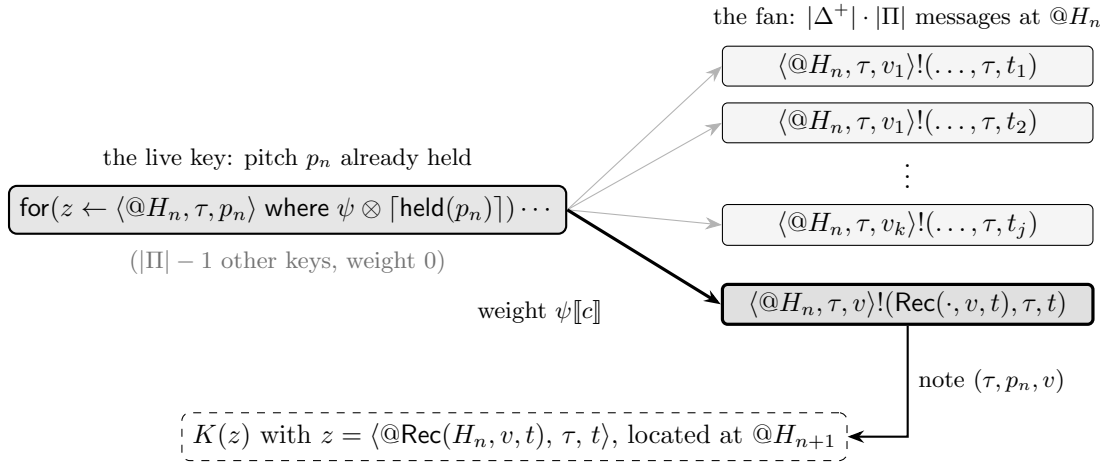


Figure 1: One step of a voice. The keys and the fan share the location $@H_n$, the quote of the previous note's record. The fan is the same for every clause; the clause weights its members and keeps exactly one key live. The chosen payload becomes the next location, so everything unchosen is left where no receipt of this voice will be.

5.2 Unbounded voices by reflection

The continuation K must bring about the next step. It cannot contain it, since that would make the term infinite; it asks for it. A per-voice *server* answers the request, and the server re-creates itself from its own code, which it keeps on a code channel, re-sending the code each time it reads it. Fix two bases B_S and B_C for the voice.

$$\begin{aligned} \text{Req}(z) &:= \langle @B_S, \tau, \epsilon \rangle!(*z, -, -) \\ \text{Server} &:= \text{for}(w \leftarrow \langle @B_S, \tau, r \rangle) (\text{Step}_{\text{Req}}(w) \mid \text{Refresh}) \\ \text{Refresh} &:= \text{for}(c \leftarrow \langle @B_C, \tau, r \rangle) (*c \mid \langle @B_C, \tau, \epsilon \rangle!(*c, -, -)) \\ \text{Voice}(p_0, u_0) &:= \text{Server} \mid \langle @B_C, \tau, \epsilon \rangle!(\text{Server}, -, -) \mid \langle @B_S, \tau, \epsilon \rangle!(H_0, -, -) \end{aligned}$$

A request passes exactly the general name $@*z$ of the new location: by Proposition 2.5 it is the greatest name there, and it is all the server needs, since the step reads only the location. Likewise the refresh receives $@\text{Server}$ and runs $*@\text{Server} \equiv \text{Server}$. The server's text mentions

only the fixed names built from B_S and B_C , the clause, and the alphabets. It does not mention itself: **Refresh** runs whatever code it finds, $*c$, and puts it back. So **Voice** is a finite term, and nothing beyond Definition 2.1 is used.

Proposition 5.3 (The machinery is silent). *In every run of $\text{Voice}(p_0, u_0)$:*

- (i) *the communications at $@B_S$ and $@B_C$ are null notes (τ, r, ϵ) , and each musical note is followed by exactly two of them, a request and a refresh;*
- (ii) *each has a single candidate, so it is resolved without chance and contends with nothing;*
- (iii) *each has the onset of the musical note that follows it and adds no time, so the performance consists of the musical notes alone;*
- (iv) *the voice is productive: between two musical notes there are finitely many null notes.*

Proof. The request is a co-channel message of duration ϵ at $@B_S$ and the server listens there on a channel of pitch r ; likewise at $@B_C$. At any time $@B_S$ holds at most one server receipt and one request, since a request is issued only by the live key's continuation and the server is re-created only after serving it; likewise $@B_C$ holds one refresh and one code message. The request is stamped at the end of the note that issued it, so the server fires at that time, adds $|\epsilon| = 0$, and stamps the next step with the same time; so does the refresh. $\square$

Remark 5.4 (Why a server, and why every key). Two shortcuts fail, and they explain the shape. Code carried as a message's payload cannot mention the name it will arrive on, because that name's location is the quote of the payload itself, and a finite term cannot contain its own quote. So the name must be handed to the code, which is what the request does; wildcards make the handover exact, since the request passes $@*z$, the typed round trip of Proposition 2.5, where a calculus without wildcards would have to invent a decoration for the server to ignore. And a name handed over by a message takes its datum from that message, which the sender can only write literally or leave open; a key on $\langle @*w, \tau, - \rangle$ has no polarity and cannot play. So the step still cannot build the one key holding the recorded pitch, and instead offers a key for every pitch and lets the clause keep one live. Wildcards dissolve the first cost and not the second. Both are paid in null notes and inert receipts, and neither is heard.

5.3 Freshness: the unplayed is inert

Along a run write $H_{n+1} := \langle @H_n, \tau, \delta_n \rangle! (H_n, \tau, p_{n+1})$, where (p_n, δ_n) is the n -th note and p_{n+1} the pitch it carried.

Lemma 5.5 (Freshness). *(i) Along one run the locations $@H_0, @H_1, \dots$ are pairwise inequivalent, and inequivalent to the voice's bases. (ii) If two voices have inequivalent bases, no location reached by one is equivalent to one reached by the other, whatever choices are made.*

Proof. Up to the unit law a record is a single message, and congruence identifies a single message only with one whose subject and payload are congruent. So $H_n \equiv H'_m$ with $n \leq m$ unwinds to $H_0 \equiv H'_{m-n}$. If $m > n$ the right side's payload is a record while H_0 's is a base, a receipt; if the bases differ the unwinding ends on two inequivalent receipts. Bases are receipts, records are messages. $\square$

Lemma 5.6 (The unplayed is inert). *In every configuration reachable from a voice, among the record locations only the current one, $@H_n$, holds a candidate of positive weight, and all of its candidates belong to the live key. The keys and messages left at earlier locations never take part in a communication.*

Proof. Each musical note consumes the live key and one fan message at $@H_n$, and the next step is placed at $@H_{n+1}$; by Lemma 5.5 nothing is ever placed at an earlier location. The other keys at $@H_n$ carry a crisp factor that reads the fixed location and is false there. $\square$

The leftovers are the continuations that were not played: inert, and a record of what was chosen against. More importantly for what follows, *the location of a voice is its past*. H_n contains every note the voice has played, nested, the most recent outermost. A spatial formula under the name predicate can therefore read the recent past to any finite depth, and this is the basis of §7.

The key's continuation cannot depend on which message wins except through the name it receives, so the fan cannot be tailored and must offer all of $\Delta^+ \times \Pi$; the transitions a clause forbids have weight 0. The process topology of a voice is the same for every clause: *the musical content is the clause*.

6 Distributions

6.1 The law of a voice

For a candidate at $@H_n$ write $\pi = p_n$ for the held pitch, δ for its duration, κ for its carried pitch, and h_n for the past recorded in H_n .

Theorem 6.1 (The law of a voice). *Let a voice run with clause ψ and suppose that at each location some candidate has positive weight. Conditionally on the past, the n -th fitting emits (p_n, v) and carries t with probability*

$$\frac{\psi[(p_n, v, t, h_n)]}{\sum_{v', t'} \psi[(p_n, v', t', h_n)]}.$$

Proof. Invariant: after n musical notes and the null notes that follow them, the configuration is congruent to $\text{Step}_{\text{Req}}(@H_n) \mid \text{Server} \mid C \mid G_n$, with C the code message and G_n inert at earlier locations. By Lemma 5.6 the only musical contention set is the star of the live key at $@H_n$, whose maximal matchings are single candidates, one per (v, t) ; the null notes are resolved without chance (Proposition 5.3). Apply Definition 3.3. Firing passes $z = \langle @Rec(\langle @H_n, \tau, p_n \rangle, v, t), \tau, t \rangle =_{\mathbf{N}} \langle @H_{n+1}, \tau, t \rangle$; the request and refresh then re-establish the invariant at $n + 1$, and the other keys and fan messages join G_{n+1} . $\square$

Example 6.2 (Markov idioms). With weighted state machines w_{Π} on pitches and w_{Δ} on durations,

$$\psi_M := \left(\bigoplus_{(s, t)} [\text{pitch}(s) \wedge \text{carry}(t)] \otimes w_{\Pi}(s, t) \right) \otimes \left(\bigoplus_{(u, v)} [\text{prev}(u) \wedge \text{dur}(v)] \otimes w_{\Delta}(u, v) \right),$$

where $\text{prev}(u) := \text{at}(\langle -, -, u \rangle! \top)$ reads the last duration from the location. By Theorem 6.1 and the factorisation below, the pitch and duration streams are independent Markov chains with the row-normalised matrices. This is the simplest idiom; everything that follows refines it.

Remark 6.3 (The instrument as a score). F1R3Skein's streams are Example 6.2 with the player's drawn machines, and its original digit streams are the uniform complete machines resolved, factor by factor, by interval decoding of two constants [6]. The simulator confirms that this reproduces the digit streams exactly (Finding 10.10). It is one score among many.

6.2 Combining distributions

A composer rarely wants one monolithic table. The clause algebra gives three ways to combine preferences, and they have different meanings.

Product: all preferences at once. If $\psi = \psi_1 \otimes \dots \otimes \psi_k$, a candidate’s weight is the product of its factors’ weights, and by Theorem 6.1 the played distribution is the normalised pointwise product — what the machine-learning literature calls a product of experts. A continuation is likely only if every factor finds it acceptable, and any factor can veto with a 0.

Lemma 6.4 (Factorised resolution). *If a star’s candidates are indexed by $A \times B$ and $\psi[(a, b)] = f(a)g(b)$, resolving the star is the same as drawing $a \propto f$ and, independently, $b \propto g$.*

Proof. The normalised product is the product of the normalisations. $\square$

So factors that look at disjoint aspects — pitch and duration, say — are heard independently, and may be realised by separate chance sources. Factors that look at the same aspect interact: a style factor and a physicality factor on the same pitch choice reweight each other.

Sum: a mixture of styles. $\psi_1 \oplus \psi_2$ adds weights. Where the two factors have comparable total mass at a location, the result is close to a mixture that sometimes speaks one idiom and sometimes the other; scaling one summand by a constant sets the blend.

Convolution: intervals rather than pitches. A factor that depends only on the interval, $\psi_I = \bigoplus_k [\text{step}(k)] \otimes w(k)$, makes the pitch walk a sum of intervals. Away from the edges of the alphabet and with no other pitch factor, the distribution of the pitch after n fittings is the n -fold convolution of the normalised interval law with the starting point; multiplying two interval factors reweights the steps, and composing two voices’ interval laws in sequence convolves them. Most idioms are interval-shaped at least in part, and this is where the word “convolving” is literal.

6.3 Style

A style, for the purposes of a score, is a factor — or a small product of factors — over the choices a voice makes. The examples below are sketches of the *shape* such factors take, not claims to have captured the idioms.

- A rock-and-roll melodic factor might favour minor-pentatonic and blues degrees over a major harmony, with a duration factor favouring swung or straight eighths; swing needs a duration alphabet with triplet values, which the composer supplies.
- A country melodic factor might favour major-pentatonic degrees and stepwise approach to chord tones, with a duration factor favouring even subdivisions.
- Both can depend on the past through the location (§7), so “approach the chord tone from a semitone below” is a factor over the last two pitches, not a table over one.

The point is architectural. Because the style is a factor, it can be held fixed while the physicality changes, blended with another style by $\oplus$, or hardened into a rule by replacing a small weight with 0.

6.4 Physicality

Instruments make some continuations easy and others hard, and players follow their hands. A *physicality factor* weights candidates by that ease.

Melodic physicality. For a single line, ease depends on the interval and, often, on where the hand is. The interval part is an interval factor. The position part needs the past: a fretted or bowed instrument's hand position is a function of the last few notes, which the location records, so a factor under $\text{at}(\cdot)$ can prefer continuations near the current position. The simulator measures the effect of exchanging such a factor under a fixed style: a stepwise instrument and a leaping instrument give measurably different lines from the same style factor (Finding 10.4).

Vertical physicality, and why it needs coupling. Closed-voice stacked thirds lie naturally under a pianist's hand, while across adjacent strings tuned in fifths they are awkward to impossible, and open voicings built on fifths lie naturally instead. A preference of this kind relates *simultaneous* notes. Independent voices cannot express it: by Theorem 8.1 their choices are independent. That is correct musically, since a hand is not a set of independent fingers, and it means that a hand's vertical physicality must be written as a coupling. The graded join of Paper II supplies it.

Definition 6.5 (Chords as joins). A *chord receipt* for $(y_1 \leftarrow x_1 \& \cdots \& y_k \leftarrow x_k \text{ where } \psi)P$, with $x_1, \dots, x_k$ channels at one location, fires only when each pattern is matched by a distinct co-channel message at that location. It emits the k notes of its patterns at one onset, and its continuation's receipts are stamped at the onset plus the longest of the k durations. Its candidates are the injective assignments of messages to patterns, and its clause sees all of them together: atoms $\text{pitch}_i, \text{dur}_i, \text{carry}_i$ for each pattern.

A chord receipt is one receipt, so Proposition 4.1 does not bite: its contention set is a star and its alternatives are whole voicings. A piano factor can then weight the voicing $(\kappa_1, \dots, \kappa_k)$ of the *next* chord by its span and by the intervals between adjacent chord tones, favouring closed thirds within a hand's reach; a factor for strings tuned in fifths can favour tones that fall on adjacent strings at the same stop. Every other factor of the score is unaffected.

Imagined physicalities. Nothing requires a physicality factor to describe a human. A hand spanning three octaves, strings tuned in thirds, or a factor that is uniform except for a vetoed interval are all one-line changes to the score, and each gives the same style a new idiom.

7 Motifs and idioms as formulae

The grading says how strongly; the formulae say about what. They carry motifs and idioms as much as the weights do.

7.1 The past is readable

Since the location is the past, the name predicate reads it. Define, over pitch sequences m ,

$$\text{Last}(\varepsilon) := \top, \quad \text{Last}(m \cdot a) := \langle -, -, - \rangle! (\text{Last}(m), -, a),$$

a formula of depth $|m|$ built from the output connective: the outermost record carries a and its payload satisfies $\text{Last}(m)$. Then $\text{at}(\text{Last}(m))$ holds exactly when the most recent pitches of the voice, ending with the held pitch, are m . Durations are read the same way from the records' subjects.

Example 7.1 (Completing a motif). To encourage a voice that has played the head of a motif $m \cdot a$ to finish it with b , multiply its clause by

$$(\lceil \text{at}(\text{Last}(m \cdot a)) \wedge \text{carry}(b) \rceil \otimes w) \oplus 1.$$

Every candidate keeps its weight except the completion, which is multiplied by $1 + w$. Motif transformations are formulae relating the carried pitch to the recorded ones: an answer transposed by k is $\lceil \text{at}(\text{Last}(m \cdot a)) \wedge \text{carry}(a + k) \rceil$, an inversion negates the intervals, and so on.

7.2 Enforcing an idiom

A crisp factor is a rule. If β is a crisp formula over the view of Definition 3.1, then under $\psi \otimes \lceil \beta \rceil$ no candidate violating β is ever played. Rules can conflict with the rest of the score, and a voice with no positive candidate falls silent for good. A behavioural modality prevents that.

Proposition 7.2 (Enforcement without deadlock). *Let β be a crisp condition on candidates, let K range over the one-hole contexts supplied by the rest of the score, and let*

$$\text{Viable}_\beta := \nu X. \bigvee_K \langle K \rangle_\beta X,$$

where $\langle K \rangle_\beta X$ holds when some β -satisfying candidate leads, in context K , to a configuration satisfying X . If the initial configuration satisfies Viable_β , then a voice with clause $\psi \otimes \lceil \beta \rceil \otimes \lceil \text{leads}(\text{Viable}_\beta) \rceil$, where $\text{leads}(\gamma)$ holds of a candidate whose residue satisfies γ , (i) never plays a note violating β and (ii) never falls silent, provided ψ is positive on some β -satisfying candidate at every step.

Proof. (i) is immediate from the crisp factor. For (ii), Viable_β is a post-fixed point: every configuration satisfying it has a β -candidate whose residue satisfies it again. The clause admits exactly such candidates, so viability is an invariant of every run, and an invariant configuration always has a positive candidate. $\square$

The greatest fixed point is in the ideal logic. It is decidable when β reads a bounded window of the past and the pitch and duration alphabets are finite, because the relevant state is then the window, the held pitch and the recorded duration — a finite graph on which ν is computed by iteration. This is the ideal-versus-checkable-fragment discipline of the rho-life notes: the ideal formula defines the idiom and the finite fragment decides it. The context labels matter: an idiom such as “a leading tone resolves up” may be viable alone and not in the company of a particular bass line, and $\langle K \rangle$ is how the formula knows the company.

7.3 Improvisation: freedom within constraint

The two layers of a clause, crisp and graded, divide the composer’s authority over an improviser cleanly.

Definition 7.3 (Admissible traces). A trace of a score is *admissible* if every note in it was emitted by a candidate of positive weight. The *support* of a score is the set of its admissible traces.

Proposition 7.4 (Constraint and preference are separate). *(i) The support of a score depends only on which candidates have weight zero, and not on the values of the positive weights. Changing the grading within the positive values changes how likely each admissible trace is, and never which traces are admissible.*

(ii) Multiplying a clause by a crisp factor $\lceil \beta \rceil$ can only shrink the support, and every admissible trace of the result satisfies β at every note.

(iii) *With the viability factor of Proposition 7.2, every admissible prefix extends to an admissible continuation: a performer who stays within the support can never be left with nothing admissible to play.*

Proof. (i) Definition 3.2 makes a candidate available exactly when its weight is positive, and the positive values enter only the probabilities of Definition 3.3. (ii) A crisp factor sends to zero the candidates where β fails and leaves the others' zero-or-positive status unchanged. (iii) is Proposition 7.2(ii). $\square$

So a composer writes *constraint* with crisp formulae and *tendency* with weights, and the two do not interfere. The crisp layer is the tune, the changes and the rules of the idiom: what an improviser may play. The graded layer is the style: what the idiom would rather play. Because the formulae include spatial and context-labelled modalities, the constraints can be about more than single notes. “Return to the head within eight bars” is a formula over the recorded past; “do not step on the soloist” is a formula over what a choice leads to in the company of the other players.

The same score then serves three kinds of performer. An algorithmic performer samples a trace with the probabilities the grading gives. A human improviser ignores the probabilities and chooses freely among the candidates of positive weight, and the score guarantees that whatever is chosen is admissible and, with the viability factor, that the choice never runs out. And a performer can mix the two, as an accompanist driven by the score might follow a human soloist. In each case the performance is a trace through the composition, and the composition is the set of traces it allows together with the weight it gives them.

7.4 Message passing as musical structure

Much of what is hard to notate is who plays in response to whom. Here that is the communication structure of the score, and the formulae give the responder access to what it is responding to.

Handoff: call and response, trading phrases. A caller plays a phrase and hands the line to a responder in the responder's timbre. Let R be a rendezvous location used once. A phrase has a written length k , so the caller needs no server: it is k steps written out, and the live key of the last one continues with $\text{Call}(z)$ instead of a request. Call emits the fan at R rather than at the caller's own location, addressed in the responder's timbre τ_B . The responder waits at R :

$$\begin{aligned} \text{Call}(x) &:= \prod_{v,t} \langle @R, \tau_B, v \rangle! (\text{Rec}_B(x, v, t), \tau_B, t), \\ \text{Resp} &:= \text{for}(y \leftarrow \langle @R, \tau_B, p_B \rangle \text{ where } \psi_B) \text{Req}_B(y), \end{aligned}$$

where Rec_B is Rec in timbre τ_B , p_B is the written first pitch of the response, and Req_B asks the responder's own server for its next step. A responder that listens on $\langle @R, -, p_B \rangle$ instead answers a call addressed in any timbre, and its first note sounds in the timbre the caller chose.

Proposition 7.5 (Handing over the history). *The responder's first note is (τ_B, p_B, v) for a fan message chosen by ψ_B , and from then on the responder's location has the caller's entire history as a sub-record. In particular ψ_B can read the call: on the first fitting through $\text{passes}(\cdot)$, and afterwards through $\text{at}(\cdot)$.*

Proof. The payload of each Call message is $\text{Rec}_B(x, v, t)$, whose payload is $*x$, the caller's final location. The responder's continuation is placed at the quote of that record. (The outermost record of the handoff also carries the caller's last held pitch, which was never played; formulae reading the call skip that one record.) $\square$

With Example 7.1’s transformation formulae, “answer the call a fifth higher” or “echo the call’s rhythm with new pitches” are clauses. Trading phrases between several players is a chain of rendezvous, and because the history is handed on, each player can respond to the whole exchange so far. In this calculus the handoff is a note — every communication is — so passing the line is itself musical: the responder’s first note is where the line changes hands.

Alternation: hockey and interlocking. A single line can alternate between two players. Give each player a server whose step uses that player’s clause and timbre, and let each player’s live key address its request to the *other* player’s server:

$$\begin{aligned}\text{Server}_A &:= \text{for}(w \leftarrow \langle @B_{S_A}, \tau_A, r \rangle) (\text{Step}_{\text{Req}_B}^{\psi_A, \tau_A}(w) \mid \text{Refresh}_A), \\ \text{Server}_B &:= \text{for}(w \leftarrow \langle @B_{S_B}, \tau_B, r \rangle) (\text{Step}_{\text{Req}_A}^{\psi_B, \tau_B}(w) \mid \text{Refresh}_B).\end{aligned}$$

The line is one line, so its interval and motif factors see every note, while each player’s clause adds that player’s own preferences; the players’ notes interleave note for note. This is the shape of interlocking in which two parts alternate strokes to form one resultant line, as with players sharing an amadinda xylophone, where $\tau_A = \tau_B$. Nothing requires the timbres to be equal: a step builds its keys and fan from the location alone, in its own player’s timbre, and the request passes the general name $@*z$, so the line can pass between instruments note by note.

Cross-rhythm: independence with a composer’s alphabet.

Example 7.6 (Three against two). Let voice A use only a triplet value of length $\frac{1}{6}$ and voice B only a length of $\frac{1}{4}$, both starting at 0 from distinct bases. Their onsets coincide exactly at multiples of $\frac{1}{2}$, whatever pitches their clauses choose, and by Theorem 8.1 the pitch choices are independent. Writing the rhythm as two independent processes, rather than as a grid with a common subdivision, keeps each part’s own pulse primary.

8 Harmony

8.1 Independence gives free harmony

Theorem 8.1 (Harmony by independence). *Let two voices have any timbres and clauses, with pairwise inequivalent bases (seed, server and code bases), and suppose neither clause mentions the other voice’s locations. Then (i) they are independent and every contention set belongs to one voice; (ii) their joint law is the product of their laws; (iii) for fixed choices every onset-ordered run has the same performance.*

Proof. (i) Lemma 5.5(ii) and Lemma 5.6; each voice’s server and code channel are at its own bases. (ii) Contention sets are resolved with independent chance and each belongs to one voice. (iii) Theorem 2.14. $\square$

Nothing coordinates two such voices: each keeps its own clock in its own data, and their notes do not depend on how their events interleave. When each voice has its own chance source, (iii) holds surely.

8.2 Coupling gives voicing

Proposition 8.2 (Simultaneity by contention). *Let two keys with distinct pitches and equal stamps share a location with a fan of $n \geq 2$ duration messages and a constant clause. One resolution emits two notes at one onset whose durations are never equal, whereas independent uniform draws coincide with probability $1/n$.*

Proof. Proposition 4.1 with $m = 2$. □

Harmony between independent parts is free; voicing — the relation between simultaneous notes of one hand or one section — is a coupling, written either as shared locations or, with full control, as a chord receipt (Definition 6.5). The calculus keeps the two apart, and the composer chooses which relation holds between which parts.

9 Musicians and instruments

The traditional division between musician and instrument is already in the calculus. Pitch lives on the receipt side and duration on the message side, so a *key* — a receipt holding a pitch, waiting — belongs to the instrument, and a *touch* — a message holding a duration — belongs to the player. A note is a touch meeting a key: where the player presses, the instrument supplies the pitch; how long the player holds, the player supplies the duration. The timbre belongs to the instrument as well. A player can leave it open, touching $\langle @K, -, q \rangle$, and the key's concrete timbre is the one that sounds.

9.1 Keys that re-arm themselves

Matching looks only at locations and timbres, so a player can choose a key only by choosing its location. An instrument therefore gives every key its own location. For a keyboard P and a pitch k let $K_{P,k}$ be a closed process naming the pair, for instance $\langle @B_P, \tau_\perp, k \rangle!(0, -, -)$ for a base B_P of the keyboard, so that distinct pairs give inequivalent locations.

Definition 9.1 (Key; keyboard; the $2n$ -hands piano). For a timbre τ and a clause ψ ,

$$\begin{aligned} \text{Key}_{P,k}^{\tau,\psi} &:= \text{for}(y \leftarrow \langle @K_{P,k}, \tau, k \rangle \text{ where } \psi) (*y \mid \text{Refresh}_{P,k}^{\tau}) \\ \text{Refresh}_{P,k}^{\tau} &:= \text{for}(c \leftarrow \langle @C_{P,k,\tau}, \tau, r \rangle) (*c \mid \langle @C_{P,k,\tau}, \tau, \epsilon \rangle!(*c, -, -)) \\ \text{Installed}_{P,k}^{\tau,\psi} &:= \text{Key}_{P,k}^{\tau,\psi} \mid \langle @C_{P,k,\tau}, \tau, \epsilon \rangle!(\text{Key}_{P,k}^{\tau,\psi}, -, -) \end{aligned}$$

where $C_{P,k,\tau}$ is the key's code location. A *keyboard* is $\prod_k \text{Installed}_{P,k}^{\tau,\psi}$ over its pitches, and the *$2n$ -hands piano* is $\prod_{P=1}^n \prod_{k=1}^{88} \text{Installed}_{P,k}^{\text{piano}, [\top]}$: n keyboards of 88 keys.

When struck, a key runs the touch's payload and fetches its own text from its code location, runs it and puts it back: the refresh trick of §5.2, one null note per strike. The key's text mentions only its two fixed locations, never itself.

Proposition 9.2 (A key sounds once at a time). *At every reachable configuration, each key location holds at most one receipt of each timbre. If a key is struck at onset t with duration δ , it is re-armed at $t + |\delta|$; and if its location holds keys of that timbre only, any other touch there has onset at least $t + |\delta|$.*

Proof. A strike consumes the key's receipt; its continuation is stamped $t + |\delta|$, so the refresh fires then, as a null note, and re-creates exactly one receipt. A waiting touch meets the new receipt, and by Definition 2.11 its onset is the larger of the two stamps. □

Two touches aimed at one key therefore queue rather than overlap, which is what a physical key does.

9.2 Writing for a player

For players we write $\langle @P, k, d \rangle$ for the touch co-channel $\langle @K_{P,k}, -, d \rangle$ at key k of keyboard P , leaving the timbre to the instrument, and $\langle @P, k, d \rangle!(Q)$ for $\langle @K_{P,k}, -, d \rangle!(Q, -, -)$. A player plays phrases, so it is convenient to have *synchronous output*: a touch that lets the player go on only when its note has ended.

Definition 9.3 (Synchronous output). With A a fresh location and κ a control timbre, write $\text{Ack}_A := \langle @A, \kappa, \epsilon \rangle!(0, -, -)$ and $\text{Wait}_A^m R := \text{for}(- \leftarrow \langle @A, \kappa, r \rangle) \cdots \text{for}(- \leftarrow \langle @A, \kappa, r \rangle) R$ with m receipts. Then

$$\begin{aligned} x!(Q); R &:= x!(Q \mid \text{Ack}_A) \mid \text{Wait}_A^1 R \\ (x_1!(Q_1) \mid \cdots \mid x_m!(Q_m)); R &:= \prod_i x_i!(Q_i \mid \text{Ack}_A) \mid \text{Wait}_A^m R \end{aligned}$$

and $;$ associates to the right.

Proposition 9.4 (Synchronous output sequences exactly). *If each x_i is a touch at an installed key and the step's touches have onsets $t_1, \dots, t_m$ and durations $\delta_1, \dots, \delta_m$, then R starts at $\max_i(t_i + |\delta_i|)$. Each touch costs one null note for its acknowledgement and one for its key's refresh; nothing else is added.*

Proof. A struck key runs the touch's payload in its continuation, stamped $t_i + |\delta_i|$, so the acknowledgement is released then. The waiting receipts were stamped earlier, so each acknowledgement fires at its release time as a null note, and the last of them releases R at the maximum. Proposition 5.3's argument gives the null-note count. $\square$

Melody and harmony appear once more, now from the player's side: a chord is touches in parallel, an arpeggio the same touches joined by $;$.

Example 9.5 (An A-minor triad, then a chord, over a bass). On a 4-hands piano ($n = 2$),

$$\begin{aligned} \text{Player}_1 &:= \langle @P_1, A3, q \rangle!(0); \langle @P_1, C3, q \rangle!(0); \langle @P_1, E3, q \rangle!(0); \\ &\quad (\langle @P_1, A3, h \rangle!(0) \mid \langle @P_1, C3, h \rangle!(0) \mid \langle @P_1, E3, h \rangle!(0)); 0 \\ \text{Player}_2 &:= \langle @P_2, A2, h \rangle!(0); \langle @P_2, E2, h \rangle!(0); 0 \end{aligned}$$

and the piece is $\text{Piano}_2 \mid \text{Player}_1 \mid \text{Player}_2$. The first player arpeggiates the A-minor triad in quarters and then strikes it as a half-note chord; the second plays a bass on the other keyboard. The performance is

onset	keyboard 1	keyboard 2
0	A3 q	A2 h
$\frac{1}{4}$	C3 q	
$\frac{1}{2}$	E3 q	E2 h
$\frac{3}{4}$	A3 h, C3 h, E3 h	

The two players touch different keyboards, so they are independent and every schedule gives this performance (Theorem 2.14). Now add a third player, on the *first* keyboard, who touches A3 for a quarter at 0. Two players reach for one key at once: that is contention, and there are exactly two outcomes. If the first player gets the key, the third plays A3 at $\frac{1}{4}$ alongside C3. If the third gets it, the first player's A3 waits until $\frac{1}{4}$ (Proposition 9.2), and because the first player writes with $;$, the rest of the phrase moves a quarter later with it, chord and all. The bass is unaffected either way.

In this division the player only sends, and choice in the calculus is made at receipts. A player that improvises therefore needs receipts of its own: it listens to a lead sheet, to the band, or to a fan of phrases it offers itself, and the grading and the idiom formulae of §6 and §7 sit there. The instrument is what it is; the score constrains the player; the player's graded choices are the improvisation (§7.3).

9.3 A variation: the chimeric instrument

On a conventional synthesizer the timbre is a setting. The player selects a patch, and every note played goes to that patch until the player selects another. Nothing about how a note is played changes what it sounds like, beyond whatever the patch was built to do with velocity.

In the calculus a key location can hold several keys, one per timbre, each re-arming itself:

$$\text{Chimera}_{P,k} := \prod_{\tau \in T} \text{Installed}_{P,k}^{\tau, \psi_\tau}.$$

A touch that names a timbre meets only the key of that timbre: this is the synthesizer, with the player choosing the patch. A touch that leaves the timbre open, $\langle @K_{P,k}, -, d \rangle$, meets every one of them. They contend, and the instrument's clauses decide which timbre sounds. The clauses see the touch (Definition 3.1): its duration, and through $\text{passes}(\cdot)$ its payload, which may carry what the player has just played. So the instrument chooses its timbre at every touch, in response to how it is being played.

Proposition 9.6 (A chimera stays whole). *At every reachable configuration, each key location of a chimera holds at most one receipt of each timbre in T , and exactly one of each when no strike there is still sounding.*

Proof. A strike consumes the receipt of the winning timbre only; the others are left where they are. The winner's refresh re-creates exactly that one (Proposition 9.2). $\square$

Because the losers are never consumed and the winner always returns, contention at a chimeric key does not leave stale alternatives behind. It also means that while one timbre of a chimeric key is sounding, its other timbres are still there: a second open touch during the note is heard on one of them, so a chimeric key can sound in two timbres at once, which no single-timbre key can. The fixed key location is safe here, where the voice needed fresh ones, because nothing at a key is ever left unchosen for good.

Example 9.7 (Articulation chooses the instrument). Let every key of a keyboard be a chimera over $\{\text{vibes}, \text{strings}\}$, with

$$\psi_{\text{vibes}} := \bigoplus_d [\text{dur}(d)] \otimes v(d), \quad \psi_{\text{strings}} := \bigoplus_d [\text{dur}(d)] \otimes s(d),$$

and weights favouring vibes for short touches and strings for long ones: $v = 9, 1, 1$ and $s = 1, 1, 9$ for eighth, quarter and half. A player who leaves the timbre open and plays the A-minor arpeggio in detached eighths is heard mostly on vibes; the same player, playing it in sustained halves, is heard mostly on strings; in quarters the instrument is undecided. A player who writes **vibes** into every touch gets vibes, whatever the articulation. As the playing moves through a sequence of articulations, the timbre follows it. With crisp weights the switch is exact; with graded weights, as here, the instrument has a temperament. And because the clauses may read the payload, the choice can depend on more than the current touch: a factor under $\text{passes}(\cdot)$ can turn the strings on when the player's recent phrase states a motif.

None of this is a feature added to the instrument. It follows from three things the calculus already has: matching on subjects only, wildcards, and contention resolved by the receipts' clauses. A synthesizer is the special case in which every touch names its timbre.

10 Measurements

The reference simulator `skeinsim.py` is stdlib-only (it uses `mpmath` only for the digits of π and e in one test). It implements the calculus generically: quoted processes are hash-consed so that

location equivalence is identity; candidates come from the premises of the communication rules, matching the two subjects only, with either subject’s timbre possibly a wildcard; each firing keeps its record, and playback reads the note off the record rather than off a label; clauses are evaluated as formulae over the atoms, including spatial atoms that walk the recorded past; and contention sets, maximal matchings and onset ordering follow the definitions. The resolver knows nothing of voices or idioms. Unless stated otherwise, machines are random over a sixteen-pitch alphabet and five durations.

Empirical finding 10.1 (The unplayed is inert). One voice, 400 notes, collection off: at every step exactly one location holds both a receipt and a message; 31,600 messages, exactly $79 = 5 \cdot 16 - 1$ per note, remain at earlier locations and none ever joins a candidate.

Empirical finding 10.2 (The law of a voice). One voice with Example 6.2’s clause, 20,000 notes: over the 54 stochastic cells the largest deviation from the pitch matrix is 0.031 ($|z| \leq 2.34$) and from the duration matrix 0.009 ($|z| \leq 1.21$), with no off-machine transition; over 117 cells of the joint law of the two choices in one fitting, the largest deviation from the product is 0.030 ($|z| \leq 2.87$). The rhythm-leads mirror image using COMM_2 gives 0.052 ($|z| \leq 2.25$) and 0.010 ($|z| \leq 1.47$). All consistent with sampling error.

Empirical finding 10.3 (Recursion by reflection is silent). The reflective voice of §5.2, run against a reference implementation of the step semantics that writes the recursion in the metalanguage, with the same clause and the same chance source for musical choices: over 3,000 notes the two performances are identical, note for note and onset for onset. The reflective voice played 6,000 null notes, exactly two per note, every one a rest of length zero. Every request delivered to the server was the general name $@*z$ of the new location, with both decorations open, exactly as Proposition 2.5 and the request’s wildcard payload decoration require. Without collection, over 200 notes, at most one record location ever held a candidate (at most two locations counting the server and code channels, which fire concurrently with the next note), and exactly $15 = |\Pi| - 1$ inert keys were left behind per note.

Empirical finding 10.4 (Product of style and physicality). A fixed random style factor over twenty-one pitches (four to seven continuations each), multiplied in turn by two interval factors: a *stepwise* instrument weighting steps of at most two degrees 3 and all else 0.3, and a *leaping* one weighting intervals of four or seven degrees 3 and all else 0.3. Over 20,000 notes each, the empirical transitions match the normalised product (largest $|z|$ 2.34 and 2.76) and no transition the product forbids occurs. The same style played on the two instruments: steps of at most two degrees are 0.582 of intervals against 0.135; leaps of four or seven are 0.051 against 0.378; the mean interval is 4.91 against 6.97 degrees.

Empirical finding 10.5 (Idioms read from the past). One voice, 10,000 notes per condition, with spatial atoms that walk the recorded past. Unconstrained, the line contains 1,303 three-fold repeated pitches. With the crisp factor forbidding a pitch equal to the last two, it contains none, and the voice never falls silent. With the motif-completion factor of Example 7.1 ($w = 20$) for the motif 0, 2, 4, the rate at which the head 0, 2 is completed by 4 rises from 0.171 to 0.775, against analytic values of 0.154 and 0.792.

Empirical finding 10.6 (Two-sided fitting). Three pitch receipts and four duration messages at one location, 2,000 trials: under maximal matchings the first resolution emits 3.0 notes and nothing remains; one at a time, the first emits 1.0 and the leftovers 2.0 more, exactly as Proposition 4.1 states.

Empirical finding 10.7 (Harmony and coupling). Two voices on distinct bases, each with its own random stream, 3,000 resolutions under five schedulers (either voice first on ties, three random tie-breaks): the five performances are identical, with 216 simultaneous strikes. Two

keys sharing a location with five duration messages, 20,000 trials: every resolution emits two notes and their durations are never equal (independent draws would coincide with probability 0.2).

Empirical finding 10.8 (The 4-hands piano). Example 9.5, with keys that re-arm by refresh and players written with synchronous output: over 20 random schedules the two-player piece has one performance, exactly the table of the example, with 16 null notes, one acknowledgement and one refresh per touch. With the third player added, the 20 schedules produce exactly two performances, the two described in the example.

Empirical finding 10.9 (The chimeric instrument). Example 9.7, the A-minor arpeggio repeated for 4,500 touches per condition: with the timbre left open, the share played on vibes is 0.904 in eighths, 0.494 in quarters and 0.091 in halves, against 0.9, 0.5 and 0.1 from the weights; with vibes written into every touch it is 1.000. No key location ever held more than two receipts.

Empirical finding 10.10 (The instrument as a score). Uniform complete machines resolved by interval decoding, pitch from the hexadecimal digits of π and duration from the base-5 digits of e : for 400 notes the streams equal the digit streams and each draw consumes exactly one digit.

11 Open threads

1. **Tempo.** Time is already in the data (Remark 2.12), so tempo can enter as a second clause, valued in a second resolution algebra, scaling $|\delta|$ when a continuation is stamped: rubato per voice, a shared tempo over a shared namespace.
2. **A surface syntax for composers.** The calculus is a core. A composer needs named factors, a library of styles and physicalities, and a way to write phrases literally where nothing should be drawn.
3. **Learning styles.** Paper II's gradient in the fibre applies directly: the weights of a style factor can be fitted to a corpus, and the formulae chosen by the composer say which features the learner may use.
4. **Orchestration as grading.** Because the payload's decoration is data and not part of matching, a fan message can hand the next note to a different timbre. Extending the fan over timbres and the candidate view with the carried timbre would make orchestration a clause factor like style and physicality; its idioms are not explored here.
5. **Joins in full.** Definition 6.5 fixes one convention for when a chord's continuation starts; others (earliest release, per-voice continuations) give different idioms and deserve comparison.
6. **Productivity in general.** A voice plays two null notes per musical note (Proposition 5.3). An arbitrary score can loop through null notes forever at one instant. A syntactic check that every cycle of null notes passes through a note of positive length would rule this out; it is not given here.
7. **Complex grading.** Over $\mathbb{C}$, two melodic paths to the same configuration would interfere. A voice already has what Paper I's quantum fragment needs — a single linear key and locations that never repeat — so audible interference between phrasings is a live question.

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